Contents

1 Introduction to Python

Python is a high-level interpreted, open-source programming language that is widely used for software development, automation, data analysis, artificial intelligence, machine learning, web development, scientific computing, and many other domains.

2 Roles in Analytics

According to the uploaded material, common analytics role include:

- Data Engineer
- Data Analyst/ Business Analyst
- Data Scientist/ Decision Scientist
- Machine Learning/ Artificial Intelligence Engineer
- MLOps (Machine Learning Operations)

3 Types of Analytics

- Descriptive Analytics
- Diagnostic Analytics
- Inferential Analytics
- Predictive Analytics
- Prescriptive Analytics
- Cognitive Analytics

4 Jupyter Notebook

Jupyter Notebook is an interactive development environment that allows users to combine executable Python code, formatted text, mathematical equations, tables, and visualizations in a single document.

4.1 Jupyter Notebook State

Every notebook cell has two states.

4.1.1 Edit State

Used for editing the contents of a cell.
Common shortcuts:

- Ctrl+Enter: Execute current cell and remain in the same cell.
- Shift+Enter: Execute current cell and move to the next cell.

4.1.2 Select State

Used for notebook management.
Important shortcuts:

- ESC+C: Copy cell
- ESC+X: Cut cell
- ESC+V: Paste cell

- ESC+A: Insert cell above
- ESC+B: Insert cell below
- ESC+DD: Delete current cell

4.2 Jupyter Notebook Modes

Three notebook modes are introduced.

- **Code Mode:** Used for writing executable Python code.
  Shortcut: ESC+Y
- **RawNBConvert Mode:** Behaves like plain text without formatting.
  Shortcut: ESC+R
- **Markdown Mode:** Used to create documentation.
  Shortcut: ESC+M

5 Comments

Comments are not executed by the Python interpreter. Primary uses include:

- Documentation
- Recording observations.

5.1 Single-Line Comment

Begins with the symbol #

5.2 Multi-Line String

Python does not have true multi-line comments.
Instead, triple-quoted strings are commonly used for documentation (docstrings).
The uploaded code is reproduced exactly below.
This is the comment
””” This is
multi-line string”””

6 Variable

A variable acts as a container that stores a value in memory.

7 Built-in Date Types

Data Type	Description	Conversion Function
int	Whole numbers	int()
float	Decimal numbers	float()
str	Sequence of characters	str()
bool	Logical values	bool()

7.1 Checking Data Types

type(a)-Returns the type of the object.

7.2 Implicit Type Conversion

a=20.2
b=10
c=a+b # Implicit conversion of b to float type
type(c)-float

8 Print

The function print() prints its argument but returns None.

9 Operators

9.1 Arithmetic Operators

- +
- -
- *
- /
- %
- //

9.2 Arithmetic Assignment Operators

- +=
- -=
- *=
- / =

9.3 Logical Operators

- and
- or
- not

9.4 Comparison Operators

- >
- <
- >=
- <=
- ==
- !=
- in

9.5 Questions

- Identify if the given number is positive (true) or negative (false).

10 Conditional Statements

Conditional Statements allow a program to make decisions based on one or more logical conditions. They control the execution flow by executing blocks of code depending on whether a condition evaluates or True or False.

Types of Conditional Statements

10.1 Type 1: if

if condition:
... statement
The block executes only when the condition is True.

10.2 Type 2: if-else

if condition:
... statement
else:
... statement
Exactly one block executes.

10.3 Type 3: if-elif-else

if condition:
... statement
elif condition:
... statement
...
else:
... statement
Python evaluates the conditions sequentially until one becomes True.

10.4 Type 4: Ternary Conditional Expression

A ternary expression performs conditional evaluation in a single line.
value_if_true if conditions else value_if_false

10.5 Question

- Identify if the given number is even or odd.
- Identify if the given number is
  positive even,
  negative even
  positive odd and
  negative odd number
- Given a value, identify if it's numeric or non-numeric type.
- Take a number as an input, and annotate:
  'foo' if the number is divisible by 3
  'buzz' if the number is divisible by 5
  'foobuzz' if the number is divisible by 3 and 5
  print the number a such if not divisible by either 3 or 5.
- Identify if a given number is positive or negative with ternary.
- Identify if a given number is positive or negative or 0 with ternary.
- Identify if the given number is even or odd with ternary.

11 Keywords

- try- Contains statements that may raise an exception.
- except- Executes when an exception occurs.
- finally- Executes regardless of whether an exception occurred.

12 Input()

Syntax: input(prompt)
Returns user input as a string.

13 int()

Convert a string into an integer.
Syntax: int(value)